192511384

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 **COURSE CODE:** Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

QUESTIONS: 1 **Total Marks:100**

Annexure A

Concept Mapping

Code of Conduct:

I Karnam Harshith (192511384) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Harshith

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Annexure A

Concept Mapping

Question 1
Show a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation). **(BL2)**

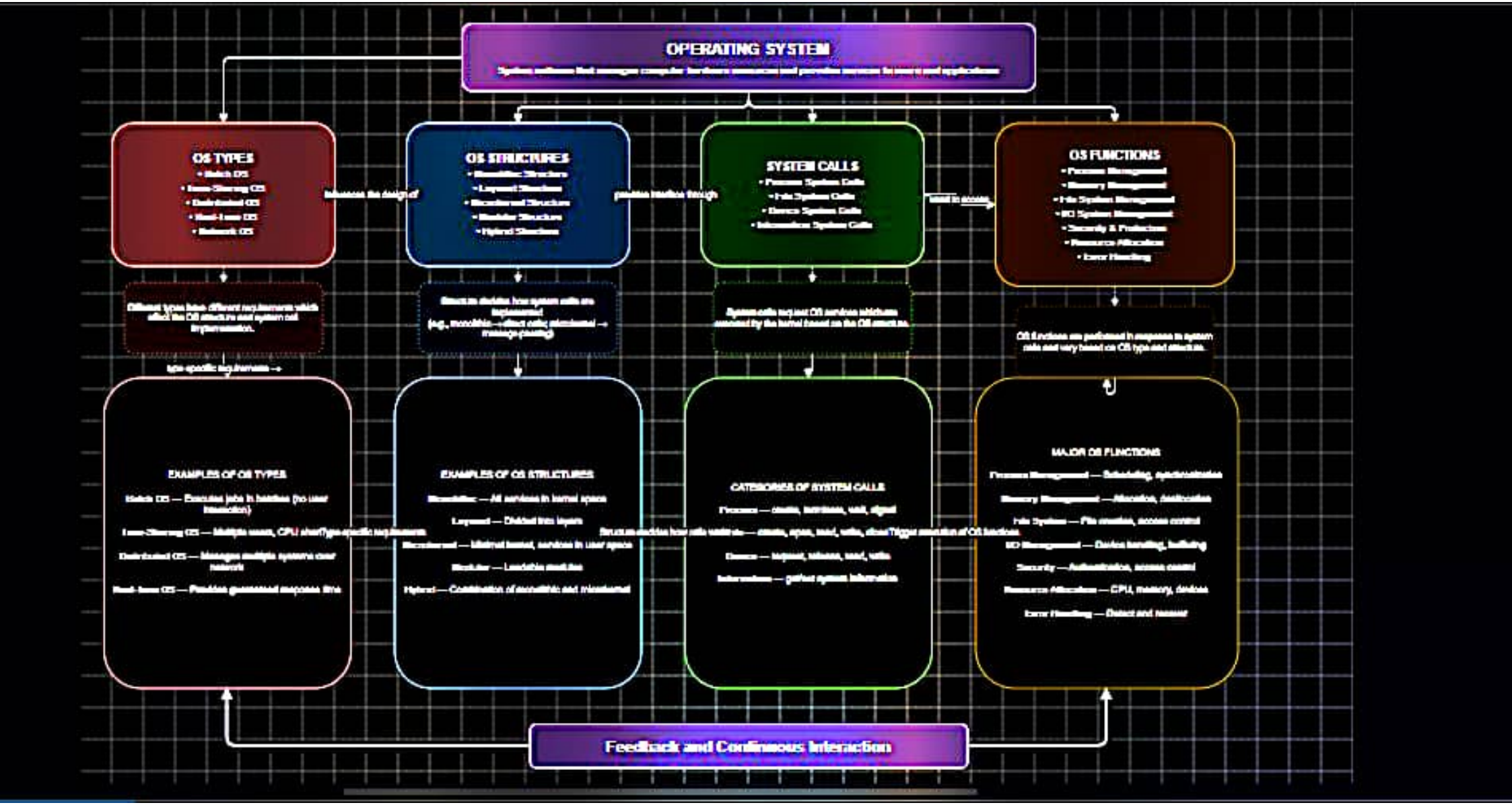
Question 2
Construct a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component. **(BL3)**

Question 3
Sketch a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases. **(BL2)**

Question 4
Illustrate a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths. **(BL3)**

Question 5
Sketch a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling. **(BL3)**

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OPERATING SYSTEMS

OS Types
- Batch OS
- Time-Sharing OS
- Distributed OS
- Real Time OS
- Network OS

Different types have different requirements which affect the OS structure and system call implementations

Examples of OS Types:
- Batch OS
- Time Sharing OS
- Distributed OS
- Real time OS

OS Structures
- Monolithic Structure
- Layered structure
- Microkernel Structure
- Modular Structure
- Hybrid Structure

Structure decides how system calls are implemented (e.g. monolithic → direct calls. microkernel → message passing)

Examples of OS Structures:
- Monolithic
- Layered
- Microkernel
- Modular
- Hybrid

System Calls
- Process System Calls
- File System Calls
- Device System Calls
- Information System calls

System calls request OS services which are executed by Kernel based on the OS structure

Categories of System Calls:
- Process System Calls
- File system calls
- Device System Calls
- Information System calls

OS Functions
- Process management
- Memory management
- File System Management
- I/O System Management
- Security & protection
- Resource allocation

OS functions are performed in response to system calls and vary based on OS type and structure

Major OS Functions:
- Process Management
- Memory Management
- File System Management
- I/O System Management

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